❸ Home Tracker: Construction Manager (CM) and Debris and Demolition Manager (DDM)

Instructions:

- Put a “B” when it is built, “S” when in stock, and “D” when it is demolished. Write the *B year* and *D year* and calculate the *Lifespan = D year – B year*.
- At the bottom, put the count of SF and MF homes Built (B), Demolished (D) that year.
- For Stock (S) *calc*: Calculate the $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$. Note that $Stock_{1999} = 0$.
- For Stock (S) *count:* put the count of SF and MF homes that are new or already existing Stock (S) that year. Compare your count to the calculation.

Address	Type	2000	2010	2020	2030	2040	2050	2060	2070	2080	B year	D year	Lifespan
1 Sweet Street	SF												
2 Sweet Street	SF												
3 Sweet Street	SF												
4 Sweet Street	SF												
5 Sweet Street	SF												
6 Sweet Street	SF												
1-3 Sugar Ave	MF												
5-7 Sugar Ave	MF												
2-4 Sugar Ave	MF												
6-8 Sugar Ave	MF												

Built (B)	SF									
Demolish (D)	SF									
Stock (S) *calc.*	SF									
Stock (S) *count*	SF									
Built (B)	MF									
Demolish (D)	MF									
Stock (S) *calc.*	MF									
Stock (S) *count*	MF									
Stock (S) *sum*	Total									

❶ Flow Tracker: Materials Warehouse Manager (MWM)

Instructions:

- Flow In_{2000} = initial new materials
- Flow $In_{>2000}$ = purchase of recycled materials from WRM
- Flow Out = sales of materials to CRM
- $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$
  - rearrange the above formula to calculate stock change based on Flow In and Flow Out
- Stock Δ = stock change = $Stock_t - Stock_{t-1}$
- $Stock_{1999} = 0$

	Brick				Wood				Metal				Glass			
Year	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock
2000																
2010																
2020																
2030																
2040																
2050																
2060																
2070																
2080																
Total																

❻ Flow Tracker: Waste & Recycling Manager (WRM)

Instructions:

- Flow In = scrap and debris from DDM
- Flow Out = sales of reusable or recycled materials to MWM
- $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$
  - rearrange the above formula to calculate stock change based on Flow In and Flow Out
- Stock Δ = stock change = $Stock_t - Stock_{t-1}$
- $Stock_{1999} = 0$

	Brick for Reuse				Wood for Reuse				Metal for Recycling				Combined Scrap & Debris for Landfill			
Year	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock
2000																
2010																
2020																
2030																
2040																
2050																
2060																
2070																
2080																
Total																

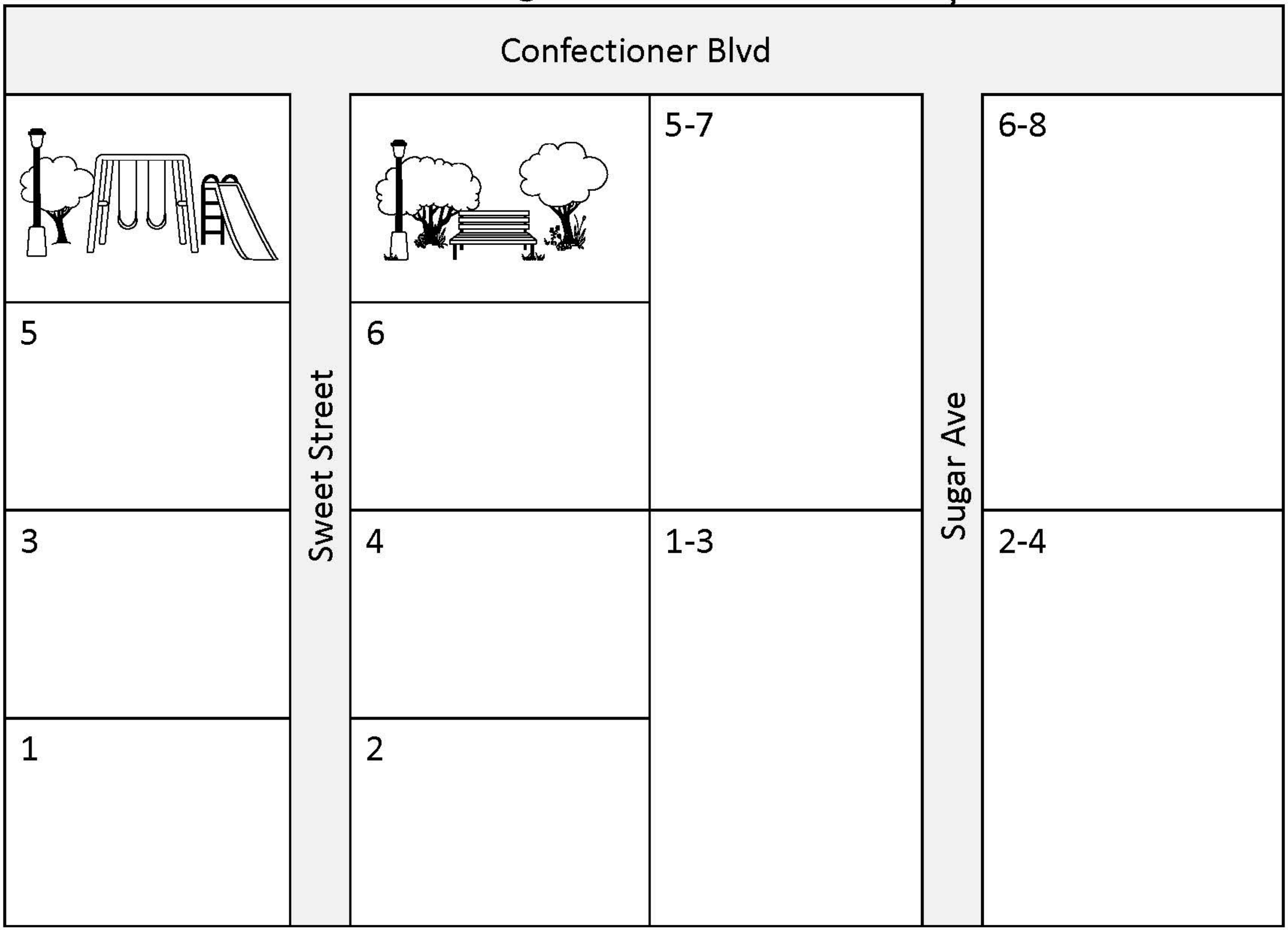
❹ Flow Tracker: Construction Manager (CM) and Debris and Demolition Manager (DDM)

Instructions:

- Flow In = construction materials from MWM
- Flow Out = scrap and debris to WRM
- $Stock_t = Stock_{t-1} + \sum FlowIn_t - \sum FlowOut_t$
    - rearrange the above formula to calculate stock change based on Flow In and Flow Out
- Stock Δ = stock change = $Stock_t - Stock_{t-1}$
- $Stock_{1999} = 0$
- For the 50th Anniversary Golden Jubilee, put the count in the specified row for comparison.

	Brick				Wood				Metal				Glass			
Year	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock	Flow In	Flow Out	Stock Δ	Stock
2000																
2010																
2020																
2030																
2040																
2050																
2050	Golden Jubilee Count:				Golden Jubilee Count:				Golden Jubilee Count:				Golden Jubilee Count:			
2060																
2070																
2080																
Total																

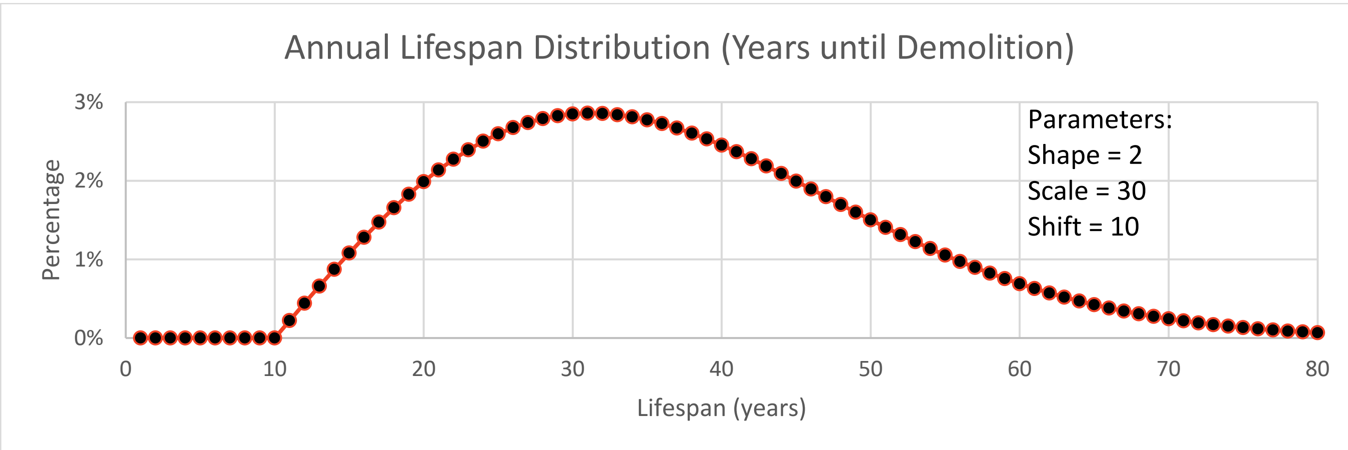
Candy Town Map



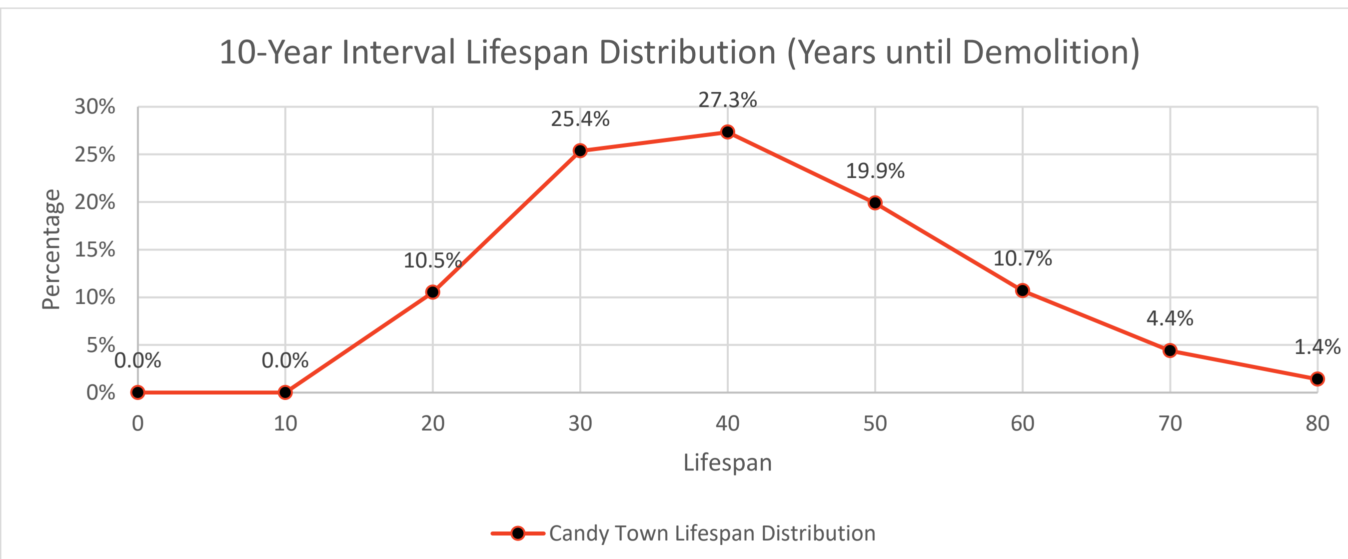
❺ Lifespan Distribution

Industrial Ecology researchers who recently settled in Hummus Town realized that the buildings in Candy Town could make an excellent case study. They modeled the lifespan distribution of the buildings using a Weibull Distribution, which is a commonly used approach for lifespans, since it was developed with that sort of purpose in mind. Instead of mean and standard deviation, the parameters for a Weibull Distribution are shape, scale, (and optionally shift).

Each point in the chart below represents the likelihood a house would be demolished after that many years.



Let's compare this to the data you collected. Since data was only available in Candy Town every decade, the researchers took the difference between *cumulative* Weibull distribution above over the preceding range. For instance, the 10.5% is the difference between the cumulative distribution at year 20 and year 10; 25.4% is the value at year 30 minus year 20.



Based on your results in the Home Tracker ❸, fill out the table below, and plot your percentages above!

- Count how many buildings have each lifespan (years).
- Find the percent of buildings with that lifespan, noting there are 10 buildings total.
- Plot your results and compare to the model estimates, plotted above and in the bottom row of the table.

Lifespan	0	10	20	30	40	50	60	70	80
Count									
Percent									
Model	0.0%	0.0%	10.5%	25.4%	27.3%	19.9%	10.7%	4.4%	1.4%

❷ Materials Warehouse Manager (MWM) Material Storage

Brick	Wood
Metal	**Glass**

❼ Waste & Recycling Manager (WRM) Material Storage

Brick for Reuse	**Wood for Reuse**
Metal for Recycling	**Combined Scrap & Debris for Landfill**